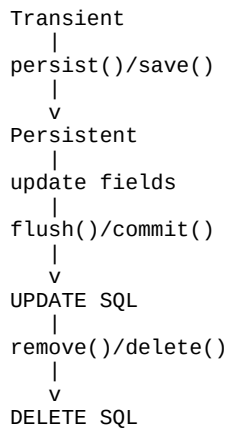


Chapter 8 - Hibernate CRUD Methods

1. Introduction

CRUD stands for Create, Read, Update and Delete. Hibernate provides several APIs for these operations. Some are Hibernate-specific and others are defined by JPA.

2. CRUD Flow



Method	Purpose	Standard
save()	Insert new entity	Hibernate
persist()	Insert new entity	JPA
get()	Read by primary key	Hibernate
getReference()	Lazy reference	JPA
update()	Reattach detached entity	Hibernate
merge()	Merge detached state	JPA
saveOrUpdate()	Insert or update	Hibernate
remove()	Delete entity	JPA
delete()	Delete entity	Hibernate
refresh()	Reload from DB	JPA/Hibernate
lock()	Reassociate/locking	Hibernate

3. save() vs persist()

Both are used to make a transient entity persistent. `persist()` is the JPA standard. `save()` returns the generated identifier, while `persist()` returns void.

4. update() vs merge()

`update()` reattaches the same detached object and may fail if another managed instance with the same identifier already exists. `merge()` copies the detached state into a managed entity and returns that managed entity.

5. saveOrUpdate()

saveOrUpdate() inserts a new entity if it has no identifier or updates an existing detached entity. It is Hibernate-specific.

6. remove() vs delete()

Both delete an entity. remove() is the JPA API. delete() is the Hibernate API.

7. refresh()

refresh() discards in-memory changes and reloads the entity state from the database.

8. lock()

lock() can reassociate an entity with the Session or apply a locking mode depending on the API used.

9. SQL Generated

```
persist()/save()  
INSERT INTO EMPLOYEE(...)  
  
update()/merge()  
UPDATE EMPLOYEE SET NAME=? WHERE ID=?  
  
remove()/delete()  
DELETE FROM EMPLOYEE WHERE ID=?  
  
refresh()  
SELECT * FROM EMPLOYEE WHERE ID=?
```

10. Example Code

```
Session session=sessionFactory.openSession();  
Transaction tx=session.beginTransaction();  
  
Employee e=new Employee();  
session.persist(e);  
  
e.setName("Siva");  
  
tx.commit();  
session.close();
```

11. Best Practices

- Prefer JPA APIs (persist, merge, remove) in new applications.
- Always perform create, update and delete operations inside a transaction.
- Use merge() when working with detached entities if another managed instance may already exist.

12. Interview Questions

1. Difference between save() and persist()? 2. Difference between update() and merge()? 3. Difference between remove() and delete()? 4. When do you use refresh()? 5. What does

saveOrUpdate() do?